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1 Introduction

To begin with, given the inherent volatility of financial markets, assessing and controlling financial risk is an essential part of investment management. Market risk, or the possibility of incurring losses as a result of swings in market prices, is one of the various types of risk that is of special importance to institutional managers and equity investors. Moreover, ARMA and GARCH models are two popular econometric methods used to forecast and quantify this risk. These models provide return and volatility forecasts that are essential for calculating Value-at-Risk. This report focuses on the econometric analysis of two major defense sector stocks: L3Harris Technologies and Northrop Grumman. Using time series techniques, a model and forecast of stock returns is done and it is estimated VaR at various confidence levels using historical daily data from the last five years, from April 2, 2020 to April 1, 2025. Prior to doing the necessary statistical analysis, the study is structured to investigate the historical behaviour of the two time series. The data is then fitted to ARMA and GARCH models in order to capture volatility clustering and other stylised characteristics that are frequently seen in financial returns. Out-of-sample predictions are produced using the fitted models, and multiple VaR measures are computed under various return distribution assumptions. Lastly, the study highlights the models' strengths and possible limitations in actual applications as it examines the relevance of econometric VaR modelling for financial risk management.

2 GARCH Models

2.1 Symmetric

2.1.1 GARCH(1,1)

The GARCH(1,1) model by **Bollerslev1986** extends the Autoregressive Conditional Heteroskedasticity (ARCH) model to capture more persistent volatility effects. The conditional variance at time t is $\text{var}(y_t) = \sigma_t^2$ is given by:

$$y_t = \sigma_t \cdot \varepsilon_t, \quad (1)$$

$$\sigma_t^2 = \omega + \alpha \cdot \varepsilon_{t-1}^2 + \beta \cdot \sigma_{t-1}^2. \quad (2)$$

where $\omega > 0$, $\alpha \geq 0$, and $\beta \geq 0$ (to ensure $\sigma_t^2 > 0$) are parameters to be estimated using maximum likelihood. ε_t is the error term of the model and is assumed to an assigned distribution. Furthermore it must hold that $\alpha + \beta < 1$ for the stability. Under this condition we have that ε_{t-1}^2 is a weakly stationary white noise sequence.

The GARCH(1,1) does not differentiate between positive and negative shocks and thus assigns an equal impact on the volatility. While this assumption helps simplify the model, it may not necessarily hold. For financial markets we expect to see that negative shocks have a stronger influence on the future volatility than positive shocks (leverage effect). In order to address possible asymmetry in the shocks, we consider asymmetric GARCH models (**Brooks2019**, **mcaleer_garch_lectures**).

2.2 Asymmetric

2.2.1 EGARCH(1,1)

An extension that can capture asymmetry is the Exponential Generalised Autoregressive Conditional Heteroskedasticity (EGARCH) model by **Nelson1991**. The EGARCH(1,1) model specifies the conditional variance σ_t^2 in an exponential form, ensuring it remains positive regardless of the parameters. This allows for flexibility when estimating the parameters.

$$\sigma_t^2 = \exp(\omega + \alpha \cdot |\eta_{t-1}| + \gamma \cdot \eta_{t-1} + \beta \cdot \log \sigma_{t-1}^2),$$

where η_{t-1} is the standardised residual:

$$\eta_{t-1} = \frac{\varepsilon_{t-1}}{\sqrt{\sigma_{t-1}^2}}. \quad (3)$$

and ω, α, γ and β are parameters to be estimated using maximum likelihood. Taking the exponent ensures the conditional variance being positive. The standardised residuals η_{t-1} are used to capture the effect of past shocks on the current volatility. The terms $|\eta_{t-1}|$ and η_{t-1} allow the model to incorporate both the magnitude and direction of the shocks. Parameter α captures the effect of the absolute value of the standardised residual $|\eta_{t-1}|$, representing the magnitude of the shock. The asymmetry parameter γ captures the effect of the actual value of the standardised residual η_{t-1} , representing the direction of the shock. γ being negative results the conditional variance σ_t^2 being larger for a negative shock ($\eta_{t-1} < 0$) than for a positive shock ($\eta_{t-1} > 0$) of the same magnitude. Allowing the model to capture potential leverage effects (**Brooks2019**, (**mcaleer_garch_lectures**)).

3 Distributions

3.1 Symmetric

3.1.1 Gaussian

The normal (Gaussian) distribution by (**Gauss1809**) is one of the most often used distributions in statistics due to its mathematical simplicity and the central limit theorem. This theorem states that the sum of independent identically distributed random variables tends a normal distribution regardless of the original distribution as the number of observations increases. It has the following probability density function (PDF) for a mean of zero and variance σ_t^2 :

$$p(r_t) = (2\pi\sigma_t^2)^{-1/2} \exp\left(-\frac{r_t^2}{2\sigma_t^2}\right). \quad (4)$$

and log-likelihood:

$$\log p(r_t) = -\frac{T}{2} \log(2\pi\sigma_t^2) - \frac{1}{2\sigma_t^2} \sum_{i=2}^T (r_t)^2. \quad (5)$$

This distribution is symmetric, meaning that the skewness equals 0. The kurtosis is relatively small, at only 3. This low kurtosis value suggests that the distribution has thin tails, which do not adequately account for the extreme events often observed in financial series. Therefore the use of a Gaussian distribution for financial data is prone to underestimate the probability and impact of extreme market events (**mcAleer_garch_lectures**).

3.1.2 Student-t

The Student-t distribution by **Gosset1908** is often used for modelling symmetric data that exhibits larger tails than those of the Gaussian. The degrees of parameter ν controls the shape of the distribution and allows for control over the heaviness of the tails. When the degrees of freedom parameter is larger than 2 ($\nu > 2$), the Student-t distribution has the following probability density function (PDF):

$$p(r_t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)} ((\nu-2)\pi\sigma_t^2)^{-1/2} \left(1 + \frac{r_t^2}{(\nu-2)\sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \quad (6)$$

and log-likelihood:

$$\begin{aligned} \log p(r_t) = & T \left[\log \Gamma\left(\frac{\nu+1}{2}\right) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \log(\pi(\nu-2)) \right] \\ & - \frac{1}{2} \sum_{i=2}^T \left[\log \sigma_t^2 + (\nu+1) \log \left(1 + \frac{r_t^2}{\sigma_t^2(\nu-2)}\right) \right]. \end{aligned} \quad (7)$$

For $2 < \nu \leq 3$ the skewness undefined. While the skewness equals zero for $\nu > 3$, indicating that the distribution is symmetric. for $2 < \nu \leq 4$ the kurtosis is infinitely large. While for $\nu > 4$ the skewness is given by: $3 + \frac{6}{\nu-4}$. This formula shows that the tails can be made thicker by adjusting the ν parameter. The Student-t distribution converges to the normal distribution as $\nu \rightarrow \infty$ (**mcAleer_garch_lectures**).

4 Data and Preliminary Analysis

4.1 Data Description (files, sample period, source)

In this report it is analyzed the daily adjusted prices of two significant defense-related stocks, Northrop Grumman (NOC) and L3Harris Technologies (LHX). The sample period spans the last five years, from 28.4.2020 to 28.4.2025.

4.2 Time Series Historical Overview

In terms of L3Harris Technologies' time series history (Figure 1), the price series demonstrates relatively stable behaviour in 2020 and 2022, but notable volatility emerges in early 2022, which corresponds with global supply chain disruptions, macroeconomic instability and the Federal Reserve's shift to higher interest rates. Throughout 2023, there is another period of mild volatility, which is probably a reflection of continued inflation and recessionary concerns.

Additionally, Northrop Grumman (Figure 2) showed a steady growth trend in 2020 and 2021, helped by strong order backlogs and higher defence spending. Increased geopolitical tensions, especially the war in Ukraine and NATO defence commitments, may have contributed to the sharp price fluctuations that occurred in late 2022 and early 2023.

A comparison of the two time series reveals increased volatility between early 2022 and mid-2023, which was caused by a number of significant world events. Financial markets experienced a great deal of volatility as a result of the Federal Reserve's rapid interest rate hikes in reaction to growing inflation. In the meantime, geopolitical risks increased due to the Russia-Ukraine war, which started in February 2022 and had a special impact on defence sector equities like Northrop Grumman and L3Harris. Stock prices were originally supported by hopes of higher defence spending, but this also heightened market volatility. The noted volatility in the stock performance of both corporations was also exacerbated by mounting concerns about a worldwide economic slowdown and possible recession.

4.3 Price and Log Return Analysis

Both L3Harris Technologies' (Figure 3) and Northrop Grumman's (Figure 4) daily log returns exhibit traditional features of financial return series. Particularly around late 2022 and early 2023, when macroeconomic uncertainty and geopolitical tensions are likely to be present, the return plots display periods of higher amplitude, indicating volatility clustering.

Figure 1: Daily Closing Prices of L3Harris

Figure 2: Daily Closing Prices of Northrop Grumman

4.4 Price and Log Return Analysis

Figure 3: Daily Log Returns of L3Harris

Figure 4: Daily Log Returns of Northrop Grumman

Table 1: Descriptive Statistics for Log Returns

	L3Harris	Northrop Grumman
Mean return	0.00013	0.00041
Standard deviation	0.0155	0.0154
Skewness	+0.128	-0.109
Kurtosis	6.19	8.51
Min / Max	-6.5% / +9.5%	-7.9% / +10.8%

The mean log returns are close to zero and positive, as would be anticipated for daily stock data. High kurtosis in both stocks indicate leptokurtic distributions, which are typical with fat-tailed financial returns. Moreover, in contrast to Northrop Grumman, which exhibits a slight left skew, L3Harris Technologies exhibits a slight right skew. The extreme values in min/max log returns suggest the presence of outliers, in which in both time series, L3Harris and Northrop Grumman (Figures 5 and 6), there were identified deviations from the mean, representing unusual daily returns and reflecting the impact of market conditions.

Strong autocorrelation is evident in the ACF of prices (Figure 7 and 8) for both stocks indicating that the prices are non-stationary. On the contrary, the ACF of log returns (Figure 9 and 10) rapidly falls to zero, implying white noise behaviour and no discernible autocorrelation, supporting the assumption of weak dependency in returns.

Regarding the stylised facts about financial returns that are observed in the data with regard to volatility clustering, the log return plots make this quite evident since significant changes are followed by even larger changes.

Additionally, returns show no serial correlation, as confirmed by ACF, which indicates that there is no serial autocorrelation. As expected, there is non-stationarity in prices; they follow a non-stationary process. Last but not least, the ACF and statistical moments suggest that the log returns are stationary, even if this is not examined here.

4.5 Stationarity Testing (by ADF Test)

Firstly, a time series' stationarity can be ascertained using the Augmented Dickey-Fuller test. By comparing the alternative hypothesis that the series is stationary with the null hypothesis that the series has a unit root, it tests the former. It was used the ADF test on both Northrop Grumman's and L3Harris' log price series.

Given the L3Harris results, which show a Dickey-Fuller statistic of -2.59 and a p-value of 0.33 , it concludes that it fails to reject the null hypothesis at the conventional significance levels. As is typical of financial price series, the log price series for L3Harris is non-stationary.

Moreover, with the Dickey-Fuller statistic for Northrop Grumman being -2.34 and the p-value being 0.44 , it is once more concluded to fail to reject the null hypothesis. Additionally, NOC's log pricing series is non-stationary as well.

These results align with financial theory, which often considers price levels to be integrated of order 1, which means that after differencing, as in the case of log returns, they become stationary. Therefore, instead of using price levels as the basis for future modelling and forecasting, the stationary log returns will be used.

5 Time Series Modelling

6 Time Series Modelling

6.1 ARMA Model Identification

It was initially analysed the autocorrelation and partial autocorrelation functions of L3Harris and Northrop Grumman in order to model their log returns. Next, the AIC and BIC criterion were used to compare different ARMA specifications.

Firstly, there is no strong serial correlation in the returns for L3Harris Technologies, as evidenced by the autocorrelation function (Figure 11) dropping out right after lag 0. Additionally, although not very significant, the partial autocorrelation function (Figure 12) exhibits faint but discernible spikes at lags 1 and 2. As compared to AR(1) and MA(1), ARMA(1,1) yielded one of the best BIC scores (-6903.63) and the lowest AIC (-6924.17) among the candidate models. Thus, ARMA(1,1) is the model used for LHX.

Similar to L3Harris, Northrop Grumman's autocorrelation function (Figure 13) showed no significant autocorrelation beyond lag 0, which is a typical feature of financial return series. Additionally, there are a few slight spikes in a partial autocorrelation function (Figure 14) at lag 1 and 2, suggesting the presence of short-term autoregressive structure.

When compared to alternative options such as AR(1), MA(1), AR(2), ARMA(1,1), ARMA(1,2) and ARMA(2,2), the ARMA(1,1) model showed the lowest AIC (-6924.173) and a competitive BIC. Therefore, this was the chosen model for L3Harris. Moreover, the same models were tested for Northrop Grumman and the ARMA(2,1) was the one that showed the lowest AIC (-6850.59) and a competitive BIC as well. As a result, it was selected the ARMA(2,1) model for Northrop Grumman.

Weak autocorrelation in both log return series supports the use of low-order ARMA models. In order to represent volatility using GARCH-type structures in the next steps, the chosen ARMA(1,1) for L3Harris and ARMA(2,1) for Northrop Grumman offer a solid basis for modelling.

To ascertain if GARCH modelling is suitable and to assess the presence of time-varying volatility, a residual analysis was performed after estimating ARMA models for the log return series.

The ARMA(2,1) residuals for Northrop Grumman indicate volatility clustering, or periods of greater variance, but they fluctuate about zero, according to the results of residual analysis. A common sign of ARCH/GARCH effects, conditional heteroskedasticity, is further confirmed by the plot's several spikes. The ARCH-LM test yielded a p-value of 0.0002 which is well below 0.05, a test statistic of 37.61, and degrees of freedom of 12. These results strongly reject the null hypothesis of no ARCH effects. The PACF of squared residuals has structure at lags 1-3 and around lag 10, while the ACF of squared residuals shows persistent autocorrelation at low lags. Thus, concludes that the adoption of GARCH models is justified by significant evidence of ARCH effects. For Northrop Grumman log returns, it is proposed an ARMA(2,1)-GARCH(1,1) model.

Next, the residuals from the ARMA(1,1) model for L3Harris Technologies are centered and stable, but they occasionally exhibit increases in volatility. Additionally, volatility bursts are visible, but they seem less frequent and lasting than in Northrop Grumman. Based on the test statistic of 20.93 in combination with 12 degrees of freedom 12, a p-value of 0.051 obtained from the ARCH-LM test. This is slightly higher than 0.05 and therefore the null hypothesis at the 5% level is not rejected. and While PACF indicates a modest, if weak, structure at early lags, ACF shows little autocorrelation. In this case, it was concluded that the evidence for ARCH effects in L3Harris is slightly significant and weaker. However, in order to ensure robustness and facilitate volatility modelling, a light ARMA(1,1)-GARCH(1,1) specification for L3Harris is suggested.

Finally, these are the proposed models that will be validated further through diagnostic testing and compared using different error distributions in the next steps:

6.2 Testing for ARCH Effects and GARCH Model Proposal

Table 2: Selected ARMA Models

Asset	Model	Criteria
L3Harris	ARMA(1,1)	Best AIC/BIC
Northrop Grumman	ARMA(2,1)	Best AIC/BIC

6.3 GARCH Model Estimation and Diagnostics

Following the estimation of ARMA-GARCH models for both assets' log returns, it employed a standard residual diagnostics to assess the model's adequacy.

As stated above, the model used for Northrop Grumman is ARMA(2,1)-GARCH(1,1). The standardised and squared residuals show no evidence of autocorrelation or ARCH effects, the conditional variance exhibits persistent behaviour suggesting that the model captures time-varying volatility, and all estimated parameters are statistically significant. A common problem in financial return series is non-normality, which is suggested by deviations in the tails in the normal QQ plot. Thus, the conditional mean and volatility structure are both effectively captured by the model. Although the GARCH(1,1) model performs well for Northrop Grumman, the normal distribution assumption might be overly restricting.

The model used for L3Harris Technologies is ARMA(1,1)-GARCH(1,1) and its sufficiency is confirmed by the statistically significant coefficients and the absence of any remaining autocorrelation or ARCH effects. Moderate volatility persistence is captured by the model. As with Northrop Grumman, the model's poor fit under the normal distribution is indicated by its failure on the adjusted Pearson test. Therefore, the ARMA(1,1)-GARCH(1,1) model works well even though the ARCH-LM outcome was a little weaker before. Although a t-Student distribution could be used to properly account for fat tails, it still does a good job of capturing the dynamics of the return series. Ljung-Box tests, conditional variance behaviour, and residual ACF/PACF plots all support the GARCH(1,1) specification in relation to the model selection. Additionally, both models show persistent volatility, which is in line with stylised financial return data. Furthermore, because positive and negative shocks are treated similarly in both models, no significant leverage or asymmetry is found. In conclusion, given normal error assumptions, ARMA(1,1)-GARCH(1,1) offers a robust model for both L3Harris and Northrop Grumman log returns based on diagnostics, stability, and theoretical consistency. The model diagnostics panels of the GARCH models for both assets can be visualized in figures 23 and 24.

6.4 GARCH Models with t-student distribution

In order to improve the modeling of heavy tails and extreme observations that can be often observed in financial time series, the re-estimation of the ARMA-GARCH(1,1) models will be done, using a t-student distribution for the error term. This allows for more flexibility in capturing heavy-tailed behaviour when compared to the standard normal distribution.

For Northrop Grumman, the ARMA(2,1)-GARCH(1,1) under a t-student distributions is given by:

Mean Equation:

$$r_t = 0.949 \cdot r_{t-1} + 0.040 \cdot r_{t-2} - \varepsilon_{t-1} + \varepsilon_t \quad (8)$$

Variance Equation:

$$\sigma_t^2 = 0.054 \cdot \varepsilon_{t-1}^2 + 0.913 \cdot \sigma_{t-1}^2 \quad (9)$$

The coefficients are statistically significant which indicates high volatility persistence $\alpha + \beta \approx 0.967$ and strong autoregressive behaviour. As previously established, the residual diagnostic tests confirmed the absence of remaining ARCH effect and of autocorrelation, which validates the suitability of the model.

For L3Harris, the ARMA(1,1)-GARCH(1,1) under a t-student distributions is given by:

Mean Equation:

$$r_t = 0.928 \cdot r_{t-1} - 0.939 \cdot \varepsilon_{t-1} + \varepsilon_t \quad (10)$$

Variance Equation:

$$\sigma_t^2 = 0.046 \cdot \varepsilon_{t-1}^2 + 0.925 \cdot \sigma_{t-1}^2 \quad (11)$$

As well as Northrop Grumman, L3Harris also has statistically significant parameters, however captures moderate volatility persistence ($\alpha + \beta \approx 0.971$). Once more, the residual diagnostic tests confirmed the absence of remaining ARCH effects and indicated that the residuals behave as a white noise.

The shape parameter gives an additional insight, as it shows the existence of substantial tail heaviness for Northrop Grumman ($v \approx 3.67$) and tails as well, for L3Harris ($v \approx 44.69$), although less extreme.

The use of a t-student distribution significantly improved the fit for Northrop Grumman, however only moderately enhanced the adequacy of the model for L3Harris.

6.5 Diagnostic reassessment of GARCH Models with student- t distributed errors

ARMA-GARCH models were re-estimated using t-student distribution on the error term after being previously estimated under the assumption of normally distributed errors. This allowed us to account for the heavier tails that are typically observed in financial return data. The ACF of both the standardized and squared residuals (Figure 25) is the confirmation that there is no significant autocorrelation remaining in either series, suggesting the adequacy of the GARCH models under the t-student distribution.

The ARCH LM Test for Northrop Grumman revealed a p-value of 0.000099, meaning that H_0 is rejected. Even after adjusting the distribution, there was still strong evidence of ARCH effects in the residuals, indicating the necessity of GARCH modelling. A few mild autocorrelations that support volatility persistence were still present in the ACF/PACF of Squared Residuals (Figure 26 and 27). This led to the conclusion that the t-student distribution ARMA(2,1)-GARCH(1,1) model remained highly justified. Clustering of volatility and heavy tails were effectively captured. Compared to the Gaussian version, the fit is better, especially when modelling extreme events.

Moreover, at the 5% significance level, the ARCH LM Test for L3Harris Technologies revealed a p-value of 0.05701, indicating that H_0 cannot be rejected. There is weak evidence of ARCH effects. There was no persistent conditional heteroskedasticity, as indicated by the majority of autocorrelations falling inside confidence bands with respect to its ACF/PACF of squared residuals (Figure 28 and 29). Therefore, it is possible to draw the conclusion that there is still limited support for GARCH effects under the t-student distribution. Although the ARMA(1,1)-GARCH(1,1) model with t-student distribution somewhat improves the fit, GARCH modelling is helpful for comparability and robustness but is not necessarily required.

Table 3: Model Evaluation Summary

Model	Residual ARCH?	Heavy tails captured	Model Fit Quality	Final Verdict
NOC (ARMA-GARCH-t)	Yes	Yes	Strong	Best model for volatility
LHX (ARMA-GARCH-t)	No (weak)	Yes	Adequate	Robust but optional

Regarding the interpretation of shape parameters, it may be stated that for Northrop Grumman $v \approx 3.67$, it indicates finite variance, heavy tails, and a distance from normal. Because of the extreme shocks to the defense industry stocks, it is justified to employ the t-distribution. Although less extreme, L3Harris $v \approx 4.69$ implies that it is also heavy-tailed. Less extreme return behaviour and a moderate departure from normalcy are supported. By using the t-student distribution instead, it was possible to increase model adequacy for both assets. The improvement is evidently needed for Northrop Grumman. while not strictly necessary, using the s-student distribution is helpful for ensuring robustness and identifying risk extremes for L3Harris. in conclusion, when comparing the models under normality, those under the t-student distribution of the error, are a clear improvement in suitability for Northrop Grumman and a reasonable enhancement for L3Harris.

7 Forecasting Log Returns and Volatility

7.1 Forecast Samples and Strategy

Using the estimated ARMA-GARCH(1,1) models with t-student innovations for both assets, out-of-sample forecasts for conditional volatility and log returns were generated. Forecasts were made over a selected evaluation window, and model performance was assessed through predicted behavior and comparative metrics. The rationale for using t-student distributions lies in their ability to capture fat tails, which are typical in financial return data and essential for accurate risk estimation.

10-days forecast for Northrop Grumman:

Table 4: Forecasted Mean Returns and Conditional Volatility

Horizon	Forecasted Mean Return	Forecasted Conditional Volatility (σ)
T+1	0.001709	0.03067
T+2	0.001527	0.03031
T+3	0.001522	0.02995
T+4	0.001510	0.02960
T+5	0.001498	0.02926
T+6	0.001487	0.02892
T+7	0.001475	0.02859
T+8	0.001464	0.02827
T+9	0.001453	0.02796
T+10	0.001441	0.02766

Although the Northrop Grumman mean returns forecast showed a slightly decreasing volatility trend (from ~ 0.0307 to ~ 0.0277), it remained consistently positive and stable (≈ 0.0015), suggesting that market expectations for Northrop Grumman include a gradual decline in risk. It is evident that Northrop Grumman is seen by the market as a stock with moderate expected returns and declining uncertainty, in line with a high-quality defense stock in a geopolitical environment that is stabilising.

10-days forecast for L3Harris:

Table 5: Forecasted Mean Returns and Conditional Volatility

Horizon	Forecasted Mean Return	Forecasted Conditional Volatility (σ)
T+1	-0.0001103	0.01634
T+2	-0.0000858	0.01632
T+3	-0.0000630	0.01630
T+4	-0.0000419	0.01628
T+5	-0.0000223	0.01626
T+6	-0.0000416	0.01625
T+7	0.0001271	0.01623
T+8	0.0002837	0.01621
T+9	0.0004290	0.01619
T+10	0.0005639	0.01618

Regarding L3Harris Technologies forecasts, mean returns were around zero, occasionally slightly

negative, indicating no discernible trend. At the same time, volatility was extremely low and stable ($\sim 0.0164 \rightarrow \sim 0.0162$), suggesting a calm market or a period of minimal trading activity. As is typical with stocks in a post-volatile stabilisation phase, it may be concluded that L3Harris is expected to be less risky but also less profitable in the short term. Additionally, it was estimated the GJR-GARCH(1,1) and eGARCH(1,1) models for both assets and used the following model selection criteria to compare their performance:

Table 6: Model Selection Criteria — Northrop Grumman (NOC)

Criterion	GJR-GARCH (NOC)	eGARCH (NOC)	Best
AIC	-5.715	-5.724	eGARCH
BIC	-5.678	-5.687	eGARCH
Shibata	-5.715	-5.723	eGARCH
Hannan-Quinn	-5.701	-5.710	eGARCH

Table 7: Model Selection Criteria — L3Harris Technologies (LHX)

Criterion	GJR-GARCH (LHX)	eGARCH (LHX)	Best
AIC	-5.654	-5.661	eGARCH
BIC	-5.621	-5.628	eGARCH
Shibata	-5.654	-5.661	eGARCH
Hannan-Quinn	-5.641	-5.649	eGARCH

The best in-sample fit for both stocks is given by the eGARCH(1,1) model. By all measures, it outperforms standard and GJR alternatives and can accommodate asymmetric volatility behaviour. These forecast values were generated through using the previously estimated ARMA-GARCH(1,1) models with the t-student distributions for both L3Harris and Northrop Grumman. These values emerged from the using the `ugarchforecast()` function in R, that calculate the expected return and conditional volatility.

7.2 Comparing Forecast Results

Table 8: Forecasted Mean Returns and Conditional Standard Deviation (σ) by Model

Model	Horizon	Forecasted Mean Return	Forecasted Conditional Std Dev (σ)
2*eGARCH	T+1	0.00095854	0.02551
	T+2	0.00079496	0.02526
2*GJR-GARCH	T+1	0.00101431	0.03251
	T+2	0.00086502	0.03214

A comparison between eGARCH(1,1) and GJR-GARCH(1,1) was made as GARCH-type models were applied to the log return series of Northrop Grumman (NOC). Both are estimated with the same mean equation (ARMA), but different volatility equations. Forecasts are made for the next two days (T+1 and T+2).

It is evident from the table that eGARCH forecasts lower volatility, whereas GJR-GARCH anticipates higher conditional variance. This indicates that eGARCH has performed better under calm market conditions, such as those recently observed in Northrop Grumman, and reflects its smoother performance to past shocks.

8 Additional Econometric Analysis

8.1 Mean Return Tests, Leverage Effects and Risk Premia Analysis

The forecasted log returns for Northrop Grumman are consistently positive and are statistically significant at any conventional level, meaning that the log returns are significantly different from zero on average.

The forecasted log returns for L3Harris are close to zero and are not statistically significant, meaning that the log returns of L3Harris are not significantly different from zero on average.

Table 9: Leverage Effect Detection via γ_1 Estimate

Asset	γ_1 (Estimate)	t-value	p-value	Leverage Effect?
NOC	0.1071378	2.760155	0.00577739	Yes
LHX	0.09942447	5.816820	5.99773e-09	Yes

Table 10: Evidence of Risk Premia via γ_1 Estimate

Asset	γ_1 (Estimate)	t-value	p-value	Evidence of risk premia?
NOC	0.000398	10.81	0.0000	Yes – statistically significant
LHX	0.000230	0.756	0.4494	No – not statistically significant

Both for Northrop Grumman and L3Harris, the estimated coefficients are statistically significant, which means there is strong evidence of leverage effects. In practice, this means that negative shocks lead to larger increases in volatility than positive shocks would have. While for Northrop Grumman shows strong evidence of a risk premium, due to persistently positive returns, the mean return is positive and significantly different from zero, reflecting a modest premia. On the other hand, L3Harris does not show evidence of risk premia, which means there is no evidence of suggesting detectable excess return over the period, that is, no evidence of risk-premia.

the most appropriate model is the eGARCH(1,1) model due to the statistical significance of its parameters and the model selection criteria. This model is able to capture the persistence in volatility and the asymmetric reaction to shocks. Showcasing the lowest AIC and BIC values and presenting well-behaved residuals (as shown on diagnostic checks seen in figures 30 and 31), the eGARCH(1,1) is the preferred one for its flexibility and robustness.

9 Value-at-Risk Estimation

9.1 Objective

The aim of this analysis is to compute the Value-at-Risk (VaR) for a long position of €10 million in the stocks L3Harris and Northrop Grumman. VaR estimates are obtained for two time horizons: 1 trading day and 5 trading days, and at 90%, 95%, and 99% confidence levels.

9.2 Methodology and Implementation

All calculations were conducted using R and the `rugarch` package.

9.2.1 Historical Parametric VaR

We computed the historical mean μ and standard deviation σ of the daily log returns. Assuming normality, the 1-day VaR is calculated as:

$$\text{VaR}_{1d} = \text{Position Value} \times |\mu + z_p \cdot \sigma| \quad (12)$$

where z_p is the standard normal quantile at the desired confidence level. The 5-day VaR is calculated as:

$$\text{VaR}_{5d} = \text{Position Value} \times |5\mu + z_p \cdot \sigma \cdot \sqrt{5}| \quad (13)$$

9.2.2 GARCH(1,1)

We fitted a GARCH(1,1) model with a constant mean to capture volatility clustering. The 1-day VaR is:

$$\text{VaR}_{1d} = \text{Position Value} \times |z_p \cdot \hat{\sigma}_{1d}| \quad (14)$$

For the 5-day VaR, we aggregated the squared daily forecasted volatilities as follows:

$$\text{VaR}_{5d} = \text{Position Value} \times \left| z_p \cdot \sqrt{\sum_{t=1}^5 \hat{\sigma}_t^2} \right| \quad (15)$$

9.2.3 AR(1)-GARCH(1,1) with Normal Errors

We extended the previous model to include an AR(1) structure in the mean equation. VaR calculations are similar to Model 9.2.2, but the return forecasts incorporate autocorrelation.

Where an AR model of order p (AR(p)) uses the past p lags to model the current value, an AR(p) process has the following formula:

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t \quad (16)$$

where c is a constant term, ϕ_i are the parameters of the model and ϵ_t is the error term (Tsay2010).

9.2.4 AR(1)-GARCH(1,1) with Student- t Errors

To better capture fat tails, we assumed the residuals follow a Student- t distribution with ν degrees of freedom. The VaR is computed using the t_ν quantile:

$$\text{VaR} = \text{Position Value} \times |t_\nu(p) \cdot \hat{\sigma}| \quad (17)$$

9.2.5 AR(1)-EGARCH(1,1) with Student- t Errors

This model allows for asymmetric volatility responses (leverage effects). Volatility is modeled in logarithmic form and fat tails are captured through Student- t innovations. The VaR formula remains the same as in Model 9.2.4.

subsection Interpretation of Estimated GARCH Parameters The following tables report the estimated parameters for various GARCH-type models and distributions fitted to the return series of L3Harris and Northrop Grumman.

Table 11: Estimated Parameters of GARCH Models for **L3Harris**

Model	μ	AR(1)	ω	α_1	β_1	γ_1	Shape
GARCH(1,1) Normal	-0.00	–	0.00	0.10	0.76	–	–
AR(1)-GARCH(1,1) Normal	-0.00	-0.03	0.00	0.11	0.75	–	–
AR(1)-GARCH(1,1) Student-t	0.00	-0.03	0.00	0.05	0.92	–	4.67
AR(1)-EGARCH(1,1) Student-t	0.00	-0.03	-0.18	-0.03	0.98	0.10	4.87

Table 12: Estimated Parameters of GARCH Models for **Northrop Grumman**

Model	μ	AR(1)	ω	α_1	β_1	γ_1	Shape
GARCH(1,1) Normal	-0.00	–	0.00	0.10	0.82	–	–
AR(1)-GARCH(1,1) Normal	-0.00	-0.05	0.00	0.10	0.82	–	–
AR(1)-GARCH(1,1) Student-t	-0.00	-0.02	0.00	0.04	0.94	–	3.90
AR(1)-EGARCH(1,1) Student-t	-0.00	-0.02	-0.18	-0.04	0.98	0.11	4.08

Across all models and both companies, the estimated mean return (μ) is close to zero, which is common for daily financial returns. For both firms the AR(1) coefficient shows small and negative (between -0.02 and -0.05) autocorrelation in the returns. Indicating weak but potentially mean-reverting behavior in short-term returns.

The estimated ω values in the standard GARCH models are extremely close to zero. This is consistent with the high persistence in volatility, as indicated by the large values of β_1 (between 0.75 and 0.94). For these models the volatility is explained through the variance and past shocks rather than the constant term ω .

For the EGARCH models, ω is estimated to be negative for both companies (approximately -0.18). Note that positivity constraint is no longer required for EGARCH models as discussed in Section 2.2.1. A negative ω suggest that the data shows strong volatility dynamics. The ARCH parameter α_1 captures the short-term reaction of volatility to recent shocks is higher in the normally distributed GARCH models (around 0.10 to 0.11) than those of the Student- t GARCH models (below 0.05). This can be explained by the heavier tails, which are captured by the shape parameter of the student- t distribution, absorbing some of the volatility explained by short-term shocks.

The estimated parameter β_1 reflects the persistence of volatility over time and is consistently high across all models (between 0.75 and 0.98). This indicates that the volatility clusters and decays slowly. Northrop Grumman has a slightly higher β_1 , meaning its volatility is more persistent.

In the EGARCH models, the asymmetry parameter γ_1 is positive for both firms (0.10 and 0.11 respectively), implying that negative shocks increase volatility more than a positive shock of equal magnitude would. This aligns with the leverage effect observed in equity markets. Finally, the shape parameter of the Student- t distribution lies between 3.9 and 4.9 across models and firms, where Northrop Grumman exhibits slightly heavier tails than L3Harris. These values confirm the presence of heavy tails in the return distributions of both stocks and indicate that Student- t innovations should be used to better capture the probability of extreme events.

9.3 Interpretation of Value at Risk (VaR) Estimates

The following tables report the estimated 1-day and 5-day Value at Risk (VaR) at 90%, 95%, and 99% confidence levels for both L3Harris and Northrop Grumman.

Table 13: 1-day Value at Risk (VaR) Estimates in EUR for **L3Harris**

Model	90% 1-day	95% 1-day	99% 1-day
Historical	197,640.80	253,413.00	358,032.30
GARCH(1,1) Normal	179,439.10	230,307.60	325,728.40
AR(1)-GARCH(1,1) Normal	178,929.70	229,653.80	324,803.80
AR(1)-GARCH(1,1) Student-t	247,005.00	338,919.30	573,872.40
AR(1)-EGARCH(1,1) Student-t	219,083.10	299,665.30	502,972.40

Table 14: 5-day Value at Risk (VaR) Estimates in EUR for **L3Harris**

Model	90% 5-day	95% 5-day	99% 5-day
Historical	444,436.6	569,147.0	803,082.9
GARCH(1,1) Normal	412,164.20	529,007.00	748,184.70
AR(1)-GARCH(1,1) Normal	411,521.70	528,182.30	747,018.30
AR(1)-GARCH(1,1) Student-t	550,484.80	755,328.50	1,278,954.00
AR(1)-EGARCH(1,1) Student-t	489,592.00	669,671.60	1,124,008.40

Table 15: 1-day Value at Risk (VaR) Estimates in EUR for **Northrop Grumman**

Model	90% 1-day	95% 1-day	99% 1-day
Historical	199,717.40	257,183.90	364,981.50
GARCH(1,1) Normal	196,591.00	252,321.80	356,863.50
AR(1)-GARCH(1,1) Normal	195,467.50	250,879.90	354,824.20
AR(1)-GARCH(1,1) Student-t	269,360.40	375,482.90	664,689.90
AR(1)-EGARCH(1,1) Student-t	279,578.30	388,063.50	678,723.40

Table 16: 5-day Value at Risk (VaR) Estimates in EUR for **Northrop Grumman**

Model	90% 5-day	95% 5-day	99% 5-day
Historical	438,300.10	566,799.10	807,841.90
GARCH(1,1) Normal	443,854.10	569,680.60	805,710.10
AR(1)-GARCH(1,1) Normal	441,843.70	567,100.30	802,060.80
AR(1)-GARCH(1,1) Student-t	598,856.40	834,793.70	1,477,774.10
AR(1)-EGARCH(1,1) Student-t	619,653.90	860,099.20	1,504,314.40

The VaR increases with the confidence level across all models, which is as expected since higher confidence levels correspond to more extreme potential losses. The compounding of risk over multiple trading days is clearly seen by the fact that the 5-day VaRs are consistently higher than the 1-day values for every model.

Surprisingly, the historical VaR produces higher estimates than the GARCH(1,1) and AR(1)-GARCH(1,1) models with normal errors across all confidence levels. This indicates that the unconditional volatility captured by the historical method reflects a higher level of recent risk than what the GARCH models predict through their smoothed conditional variance and may underestimate near-term risk.

The GARCH(1,1) and AR(1)-GARCH(1,1) models with normally distributed errors yield nearly identical VaR estimates. These models produce the lowest VaRs among all approaches due to the assumption of thin-tailed return distributions, which tend to underestimate the likelihood of extreme losses. The AR(1)-GARCH(1,1) model also produces slightly lower VaR estimates than the standard GARCH(1,1) model. This is due to the inclusion of a small, negative AR(1) coefficient in the mean equation, which introduces a weak mean-reverting effect that slightly reduces the expected loss.

The inclusion of Student- t innovations leads to significantly higher VaR estimates compared to the normal models, especially at the 99% confidence level, showcasing the importance of tail risk when estimating potential losses under extreme conditions. The difference is to be expected as heavier tails capture more extreme event and we have estimated low shape parameters for the Student- t distribution.

Interestingly, the EGARCH model with Student- t errors behaves differently across the two firms. For L3Harris, the AR(1)-EGARCH(1,1) Student- t model results in slightly lower VaR estimates than the symmetric Student- t GARCH model. This may be due to the return series showing fewer extreme negative shocks, allowing the asymmetry in the EGARCH model to reduce volatility. The asymmetry parameter γ_1 for L3Harris is estimated at 0.10, and the shape parameter is relatively high at 4.87, suggesting less extreme tail behavior.

For Northrop Grumman, however, the EGARCH model produces the highest VaR across all specifications. This indicates that the model captures strong asymmetry in volatility, likely driven by large negative returns. The slightly higher asymmetry parameter $\gamma_1 = 0.11$ and lower shape parameter of 4.08 further support this, suggesting heavier tails and a stronger leverage effect. As a result, the 5-day 99% VaR exceeds €1.5 million under the EGARCH specification, reflecting an aggressive risk forecast.

The substantial difference in VaR estimates show just how sensitive the results are to choices made such as the horizon, the confidence levels but most important of all to the specification of volatility dynamics and the assumed distribution of returns. Additionally a more complex model is less intuitive and harder to explain.

10 Conclusion

This report analysed the market risk of L3Harris Technologies and Northrop Grumman using GARCH-type models and Value-at-Risk estimation. The results showed clear differences between models, especially when comparing normal and Student- t innovations. Volatility clustering and persistence were present in both stocks, with Northrop Grumman showing stronger tail risk and asymmetry.

The VaR estimates varied considerably depending on the model and confidence level. Models assuming normality tended to underestimate risk, particularly at higher confidence levels. The Student- t models produced higher VaRs, capturing the heavier tails in the data.

While these models provide useful insights, they also come with limitations. Their sensitivity to assumptions, difficulty in implementation, and lack of transparency can be challenging in practice.